



SIMATS
ENGINEERING



SIMATS
Sreeveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING

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COURSE CODE: CSA0311

COURSE NAME: DATA STRUCTURES

Output Prediction

1.

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name "SIMATS Online Programming Lab", and the user's name "Murali Krishnan N" with their registration number "192524090RB". The main content area shows a question titled "Largest and Smallest element" worth 1 mark. The question asks to write a C program to find the largest and smallest element in an array. The user's code is displayed in a dark-themed editor, and the output is shown in a light box. The code defines an array of 100 integers, reads the number of elements (5) and the elements themselves (24, 26, 39, 56, 10), and then finds the largest and smallest values. The output shows the largest element as 56 and the smallest element as 10.

QUESTIONS

- Q1 Largest and Smallest element 1 mark
- Q2 Second Largest Element 1 mark
- Q3 Occurrence of an element 1 mark

2/3 answered

Q1 Largest and Smallest element 1 mark

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[100],n,i;
5     int largest,smallest;
6     printf("enter the number of elements: ");
7     scanf("%d",&n);
8     for(i=0;i<n;i++)
9     {
10         scanf("%d",&a[i]);
11     }
12     largest=smallest=a[0];
13     for(i=1;i<n;i++)
14     {
15         if(a[i]>largest)
16             largest=a[i];
17         if(a[i]<smallest)
18             smallest = a[i];
19     }
20     printf("largest element = %d\n",largest);
21     printf("smallest element= %d\n",smallest);
22     return 0;
23 }
```

Input

```
5
24 26 39 56 10
```

Output

```
enter the number of elements:
largest element = 56
smallest element= 10
```

Run Save Submit

2.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "Murali Krishnan N" with a dropdown arrow. The page is for "DS PROGRAMMING".

On the left, a "QUESTIONS" sidebar lists three questions:

- Q1: Largest and Smallest element (1 mark)
- Q2: Second Largest Element (1 mark) - This is the active question.
- Q3: Occurrence of an element (1 mark)

 It also shows "2/3 answered".

The main area for Question 2 is titled "Q2 Second Largest Element" with a value of "1 mark". The instruction is "Write a C program for Find the Second Largest Element".

The "Your Code" section contains the following C code:


```
1 #include<stdio.h>
2 int main()
3 {
4     int a[20],n,i,largest,second;
5     scanf("%d",&n);
6     for(i = 0; i < n ; i++)
7         scanf("%d",&a[i]);
8     largest = second = -9999;
9     for(i = 0; i < n ; i++)
10         if(a[i] > largest)
11         {
12             second = largest;
13             largest = a[i];
14         }
15     else if(a[i] > second && a[i] != largest)
16     {
17         second = a[i];
18     }
19     printf("%d",second);
20     return 0;
21 }
```

The "Input" field contains the text "5" and "10 25 8 40 15". The "Output" field displays "25".

At the bottom, there are buttons for "Run", "Save", and "Submit".

3.

The screenshot shows the SIMATS Online Programming Lab interface. The header is identical to the previous screenshot, showing the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user's name "Murali Krishnan N".

On the left, the "QUESTIONS" sidebar shows:

- Q1: Largest and Smallest element (1 mark)
- Q2: Second Largest Element (1 mark)
- Q3: Occurrence of an element (1 mark) - This is the active question.

 It shows "3/3 answered".

The main area for Question 3 is titled "Q3 Occurrence of an element" with a value of "1 mark". The instruction is "Write a C program for Find the First and Last Occurrence of an Element".

The "Your Code" section contains the following C code:


```
1 #include<stdio.h>
2 int main()
3 {
4     int n, key;
5     int first = -1, last = -1;
6     if(scanf("%d",&n) != 1 || n <= 0 ){
7         return 0;
8     }
9     int arr[n];
10    for ( int i = 0; i < n; i++){
11        scanf("%d", &arr[i]);
12    }
13    {
14        scanf("%d",&key);
15        for ( int i = 0; i < n; i++){
16            if (arr[i] == key ){
17                if (first == -1){
18                    first = i;
19                }last = i;
20            }
21        }
22    }
23    if (first != -1){
24        printf(" the first occurrence = %d\n ",first);
25        printf(" the second occurrence = %d\n",last);
26    }else{
27        printf("elements not found\n");
28    }
29    return 0;
30 }
```

The "Input" field contains the text "6", "1 2 3 2 4 2", and "2". The "Output" field displays "the first occurrence = 1" and "the second occurrence = 5".